# Six-Cone Partition of the Three-Sphere and Flavour Projections

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We present the mathematical proof that the three-sphere S3 is partitioned into six congruent cones with a maximum half-opening angle of 30 degrees. This geometric division represents the unique stable projection of the 24-cell onto S3. We map this partition to the flavor structure of elementary particles, showing that neutrino mixing angles are projections of this six-cone geometry.

## I. INTRODUCTION

This paper details the angular partition of spatial geometry in SDGFT. The 6-cone partition serves as the template for flavor structures, matching the three generations of quarks and leptons.

## II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\Omega_{\text{cone}} = 2\pi(1 - \cos\theta_{\text{max}}) = 2\pi\left(1 - \frac{\sqrt{3}}{2}\right) \tag{1}$$

$$N_{\text{cones}} = \frac{360^\circ}{2 \times 30^\circ} = 6 \quad \text{(exact spatial sectors)} \tag{2}$$

## III. THE 30-DEGREE BOUNDARY

The boundary angle is exactly 30 degrees, corresponding to the tree-level Weinberg angle parameter. We discuss the stability of this boundary under local deformations.

## IV. CONCLUSION

We conclude that spatial flavor is an emergent property of the three-sphere's projective geometry under 24-cell restrictions.

[1] D. A. Besemer, "Axiomatic Foundations of SDGFT," SDGFT-2026-01 (in preparation).

[2] D. A. Besemer, "Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT," SDGFT-2026-04.

[3] D. A. Besemer, `sdgft` Python package, `https://github.com/david-besemer/sdgft` (2026).

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